Analysis II: Terence Tao

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Metric Spaces

§ 1.1 Definitions and Examples

Lemma 1.1. Let $(x_n)_{n=m}^{\infty}$ be a sequence of real numbers, and let x be another real number. Then $(x_n)_{n=m}^{\infty}$ converges to x if and only if $\lim_{n\to\infty} d(x_n, x) = 0$.

We would now like to generalize this notion of convergence, so that one can take limits not just of sequences of real numbers, but also sequences of complex numbers, or sequences of vectors, or sequences of matrices, or sequences of functions, even sequences of sequences.

Definition 1.1 (Metric Spaces)**.** Am metric space (X, d) is a space X of objects (called points), together with a distance function or metric $d : X \times X \to [0, +\infty)$, which associates with each pair x, y of points in X a non-negative real number $d(x, y) \geq 0$. Furthermore, the metric must satisfy the following four axioms:

1. For any $x \in X$, we have $d(x, x) = 0$.
2. (Positivity) For any distinct $x, y \in X$, we have $d(x, y) > 0$.
3. (Symmetry) For any $x, y \in X$, we have $d(x, y) = d(y, x)$.
4. (Triangle inequality) For any $x, y, z \in X$, we have $d(x, z) \leq d(x, y) + d(y, z)$.

Example. Let $\mathbb{R}$ be the real numbers, and let $d : \mathbb{R} \times \mathbb{R} \to [0, \infty)$ be the metric $d(x, y) := |x - y|$. Then $(\mathbb{R}, d)$ is a metric space. We refer to d as the standard metric on $\mathbb{R}$, and if we refer to $\mathbb{R}$ as a metric space, we assume that the metric is given by the standard metric d unless otherwise specified.

Example (Induced metric spaces)**.** Let (X, d) be any metric space, and let Y be a subset of X. Then we can restric the metric function $d : X \times X \to [0, +\infty)$ to the subset $Y \times Y$ of $X \times X$ to create a restricted metric function $d_{|Y\times Y} : Y \times Y \to [0, +\infty)$ of Y; this is known as the metric on Y induced by the metric d on X. The pair $(Y, d_{|Y\times Y})$ is a metric space and is known as the subspace of (X, d) induced by Y. Thus for instance the metric on the real line induces a metric space structure on any subset of the reals, such as the integers $\mathbb{Z}$, or an interval $[a, b]$, etc.

Example (Euclidean spaces)**.** Let $n \geq 1$ be a natural number, and let $\mathbb{R}^n$ be the space of n-tuples of real numbers:

$$\mathbb{R}^n = \{(x_1, x_1, \ldots, x_n) \mid x_1, \ldots, x_n \in \mathbb{R}\}.$$

We define the Euclidean metric (also called the l^2 metric) $d_{l^2} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ by

$$d_{l^2}((x_1, \dots, x_n), (y_1, \dots, y_n)) := \sqrt{(x_1 - y_1)^2 + \dots + (x_n - y_n)^2}$$
$$= \left(\sum_{i=1}^{n} (x_i - y_i)^2 \right)^{\frac{1}{2}}.$$

Example (Taxi-cab metric)**.** Again let $n \geq 1$, and let $\mathbb{R}^n$ be as before. But now we use a different metric d_{l^1}, the so-called taxicab metric (or l^1 metric), defined by

$$d_{l^1}((x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n)) := |x_1 - y_1| + \dots + |x_n - y_n|$$
$$= \sum_{i=1}^{n} |x_i - y_i|.$$

The metrics above are not quite the same, but we do have the inequalities

$$d_{l^2}(x, y) \leq d_{l^1}(x, y) \leq \sqrt{n} d_{l^2}(x, y)$$

for all x, y .

Example (Sup norm metric)**.** Again let $n \geq 1$, and let $\mathbb{R}^n$ be as before. But now we use a different metric d_{l^∞}, the so-called sup norm metric (or l^∞ metric), defined by

$$d_{l^\infty}((x_1, \dots, x_n), (y_1, \dots, y_n)) := \sup\{|x_i - y_i| \mid 1 \leq i \leq n\}.$$

Example (Discrete metric)**.** Let X be an arbitrary set (finite or infinite), and define the discrete metric d_{disc} by setting $d_{\text{disc}}(x, y) := 0$ when $x = y$, and $d_{\text{disc}}(x, y) := 1$ when $x \neq y$. Thus, in this metric, all points are equally far apart. The space (X, d_{disc}) is a metric space. Thus every set X has atleast one metric on it.

Definition 1.2 (Convergence of sequences in metric spaces)**.** Let m be an integer, (X, d) be a metric space and let $(x^{(n)})_{n=m}^{\infty}$ be a sequence of points in X (i.e., for every natural number $n \geq m$, we assume that $x^{(n)}$ is an element of X). Let x be a point in X. We say that $(x^{(n)})_{n=m}^{\infty}$ converges to x w.r.t the metric d, if and only if the limit $\lim_{n \to \infty} d(x^{(n)}, x)$ exists and is equal to 0. In other words, $(x^{(n)})_{n=m}^{\infty}$ converges to x w.r.t d if and only if for every $\varepsilon > 0$, there exists an $N \geq m$ such that $d(x^{(n)}, x) \leq \varepsilon$ for all $n \geq N$.

Example. We work in the Euclidean space $\mathbb{R}^2$ with the standard Euclidean metric d_{l^2}. Let $(x^{(n)})_{n=1}^{\infty}$ denote the sequence $x^{(n)} := (1/n, 1/n)$ in $\mathbb{R}^2$, i.e., we are considering the sequence $(1, 1), (1/2, 1/2), (1/3, 1/3), \dots$. Then this sequence converges to $(0, 0)$ w.r.t the Euclidean metric d_{l^2}, since

$$\lim_{n \to \infty} d_{l^2}(x^{(n)}, (0, 0)) = \lim_{n \to \infty} \sqrt{\frac{1}{n^2} + \frac{1}{n^2}} = \lim_{n \to \infty} \frac{\sqrt{2}}{n} = 0.$$

The sequence $(x^{(n)})_{n=1}^{\infty}$ also converges to $(0,0)$ w.r.t the taxi-cab metric d_{l^1}, since

$$\lim_{n\to\infty} d_{l^1}(x^{(n)},(0,0)) = \lim_{n\to\infty} \frac{1}{n} + \frac{1}{n} = \lim_{n\to\infty} \frac{2}{n} = 0.$$

Similarly the sequence converges to $(0,0)$ in the sup norm metric d_{l^∞}. However, the sequence $(x^{(n)})_{n=1}^{\infty}$ does not converge to $(0,0)$ in the discrete metric d_{disc}, since

$$\lim_{n\to\infty} d_{\text{disc}}(x^{(n)},(0,0)) = \lim_{n\to\infty} 1 = 1 \neq 0.$$

In other words, a sequence converges in the Euclidean, taxi-cab, or sup norm metric if and only if each of its components converges individually.

Proposition 1.1 (Convergence in the discrete metric). Let X be any set, and let d_{disc} be the discrete metric on X. Let $(x^{(n)})_{n=m}^{\infty}$ be a sequence of points in X, and let x be a point in X. Then $(x^{(n)})_{n=m}^{\infty}$ converges to x w.r.t the discrete metric d_{disc} if and only if there exists an $N \geq m$ such that $x^{(n)} = x$ for all $n \geq N$.

Proposition 1.2 (Uniqueness of limits). Let (X,d) be a metric space, and let $(x^{(n)})_{n=m}^{\infty}$ be a sequence in X. Suppose that there are two points $x, x' \in X$ such that $(x^{(n)})_{n=m}^{\infty}$ converges to x w.r.t to d, and $(x^{(n)})_{n=m}^{\infty}$ also converges to x' w.r.t d. Then we have $x = x'$.

Notation. If $(x^{(n)})_{n=m}^{\infty}$ converges to x in the metric d, then we write $d - \lim_{n\to\infty} x^{(n)} = x$, or simply $\lim_{n\to\infty} x^{(n)} = x$ when there is no confusion as to what d is.

§ 1.2 Some point-set topology of metric spaces

Having defined the operation of convergence on metric spaces, we now define a couple other related notations, including that of open set, closed set, interior, exterior, boundary, and adherent point. The study of such notions is known as point-set topology.

Definition 1.3 (Balls). Let (X,d) be a metric space, let x_0 be a point in X, and let $r > 0$. We define the ball $B_{(X,d)}(x_0,r)$ in X, centered at x_0, and with radius r, in the metric d, to be the set

$$B_{(X,d)}(x_0,r) := \{x \in X \mid d(x,x_0) < r\}.$$

When it is clear what the metric space (X,d) is, we shall abbreviate $B_{(X,d)}(x_0,r)$ as just $B(x_0,r)$.

Definition 1.4 (Interior, exterior, boundary). Let (X, d) be a metric space, let E be a subset of X, and let x_0 be a point in X. We say that x_0 is an interior point of E if there exists a radius $r > 0$ such that $B(x_0, r) \subseteq E$. We say that x_0 is an exterior point of E if there exists a radius $r > 0$ such that $B(x_0, r) \cap E = \emptyset$. We say that x_0 is a boundary point of E if it is neither an interior point nor an exterior point of E.

The set of all interior points of E is called the interior of E and is sometimes denoted $\mathrm{int}(E)$. The set of exterior points of E is called the exterior of E and is sometimes denoted $\mathrm{ext}(E)$. The set of boundary points of E is called the boundary of E and is sometimes denoted ∂E.

Example. When we give a set X the discrete metric d_{disc}, and E is any subset of X, then every element of E is an interior point of E, every point not contained in E is an exterior point of E, and there are no boundary points.

Definition 1.5 (Closure). Let (X, d) be a metric space, let E be a subset of X, and let x_0 be a point in X. We say that x_0 iss an adherent point of E if for every radius $r > 0$, the ball $B(x_0, r)$ has a non-empty intersection with E. The set of all adherent points of E is called the closure of E and is denoted $\overline{E}$.

Proposition 1.3. Let (X, d) be a metric space, let E be a subset of X, and let x_0 be a point in X. Then the following statements are logically equivalent.

1. x_0 is an adherent point of E.
2. x_0 is either an interior point or a boundary point of E.
3. There exists a sequence $(x_n)_{n=1}^{\infty}$ in E which converges to x_0 w.r.t the metric d.

Corollary 1.1. Let (X, d) be a metric space, and let E be a subset of X. Then $\overline{E} = \mathrm{int}(E) \cup \partial E = X \backslash \mathrm{ext}(E)$.

Definition 1.6 (Open and closed sets). Let (X, d) be a metric space, and let E be a subset of X. We say that E is closed if it contains all of its boundary points, i.e., $\partial E \subseteq E$. We say that E is open if it contains none of its boundary points, i.e., $\partial E \cap E = \emptyset$. If E contains some of its boundary points but not others, then it is neither open nor closed.

Example. We work in the real line $\mathbb{R}$ with the standard metric d. The set $(1, 2)$ does not contain either of its boundary points $1, 2$ and is hence open. The set $[1, 2]$ contains both of its boundary points $1, 2$ and is hence closed. The set $[1, 2)$ contains one if its boundary points 1, but does not contain the other boundary point 2, so is neither open nor closed.

Proposition 1.4 (Basic properties of open and closed sets)**.** Let (X, d) be a metric space.

1. Let E be a subset of X. Then E is open if and only if $E = \text{int}(E)$. In other words, E is open if and only if for every $x \in E$, there exists an $r > 0$ such that $B(x, r) \subseteq E$.
2. Let E be a subset of X. Then E is closed if and only if E contains all its adherent points. In other words, E is closed if and only if for every convergent sequence $(x_n)_{n=m}^{\infty}$ in E, the limit $\lim_{n\to\infty} x_n$ of that sequence also lies in E.
3. For any $x_0 \in X$ and $r > 0$, then the ball $B(x_0, r)$ is an open set. The set $\{x \in X \mid d(x, x_0) \leq r\}$ is a closed set.
4. Any singleton set $\{x_0\}$, where $x_0 \in X$, is automatically closed.
5. If E is a subset of X, then E is open if and only if the complement $X \backslash E := \{x \in X \mid x \notin E\}$ is closed.
6. If $E_1, \ldots, E_n$ are a finite collection of open sets in X, then $E_1 \cap E_2 \cap \cdots \cap E_n$ is also open. If $F_1, \ldots, F_n$ is a finite collection of closed sets in X, then $F_1 \cup F_2 \cup \cdots \cup F_n$ is also closed.
7. If $\{E_\alpha\}_{\alpha\in I}$ is a collection of open sets in X (where the index set I could be finite, countable or uncountable), then the union $\bigcup_{\alpha\in I} E_\alpha := \{x \in X \mid x \in E_\alpha \text{ for some } \alpha \in I\}$ is also open. If $\{F_\alpha\}_{\alpha\in I}$ is a collection of closed sets in X, then the intersection $\bigcap_{\alpha\in I} F_\alpha := \{x \in X \mid x \in F_\alpha \text{ for all } \alpha \in I\}$ is also closed.
8. If E is any subset of X, then $\text{int}(E)$ is the largest open set which is contained in E; in other words, $\text{int}(E)$ is open, and given any other open set $V \subseteq E$, we have $V \subseteq \text{int}(E)$. Similarly $\overline{E}$ is the smallest closed set which contains E; in other words, $\overline{E}$ is closed, and given any other closed set $K \supset E, K \supset \overline{E}$.

§ 1.3 Relative topology

Example. Consider the plane $\mathbb{R}^2$ with the Euclidean metric d_{l^2}. Inside the plane, we can find the x-axis $X := \{(x, 0) \mid x \in \mathbb{R}\}$. The metric d_{l^2} can be restricted to X, creating a subspace $(X, d_{l^2|X\times X})$ of $(\mathbb{R}^2, d_{l^2})$. Now consider the set

$$E := \{(x, 0) \mid -1 < x < 1\}$$

which is both a subset of X and of $\mathbb{R}^2$. Viewed as a subset of $\mathbb{R}^2$, it is not open, because the point 0, for instance, lies in E, but is not an interior point of E. Any ball $B_{\mathbb{R}^2, d_{l^2}}(0, r)$ will contain atleast one point that lies outside of the x-axis, and hence outside of E. On the other hand, if viewed as a subset of X, it is open; every point of E is an interior point of E w.r.t the metric space $(X, d_{l^2|X\times X})$.

Definition 1.7 (Relative topology). Let (X, d) be a metric space, let Y be a subset of X, and let E be a subset of Y. We say that E is relatively open w.r.t Y if it is open in the metric subspace $(Y, d_{|Y \times Y})$. Similarly, we say that E is relatively closed w.r.t Y if it is closed in the metric space $(Y, d_{|Y \times Y})$.

Proposition 1.5. Let (X, d) be a metric space, let Y be a subset of X, and let E be a subset of Y.

1. E is relatively open w.r.t Y if and only if $E = V \cap Y$ for some set $V \subseteq X$ which is open in X.
2. E is relatively closed w.r.t Y if and only if $E = K \cap Y$ for some set $K \subseteq X$ which is closed in X.

Proof. We just prove (1). First suppose that E is relatively open w.r.t Y. Then E is open in the metric space $(Y, d_{|Y \times Y})$. Thus, for every $x \in E$, there exists a radius $r > 0$ such that the ball $B_{(Y,d_{|Y\times Y})}(x, r)$ is contained in E. The radius r depends on x; to emphasize this we write r_x instead of r, thus for every $x \in E$ the ball $B_{(Y,d_{|Y\times Y})}(x, r_x)$ is contained in E.

Now consider the set

$$V := \bigcup_{x \in E} B_{(X,d)}(x, r_x).$$

This is a subset of X and V is open. Now we prove that $E = V \cap Y$. Certainly any point $x \in E$ lies in $V \cap Y$, since it lies in Y and it also lies in $B_{(X,d)}(x, r_x)$, and hence in V. Now suppose that y is a point in $V \cap Y$. Then $y \in V$, which implies that there exists an $x \in E$ such that $y \in B_{(X,d)}(x, r_x)$. But since y is also in Y, this implies that $y \in B_{(Y,d_{|Y\times Y})}(x, r_x)$. But by definition of r_x, this means that $y \in E$, as desired.

Now we do the converse. Suppose that $E = V \cap Y$ for some open set V; we have to show that E is relatively open w.r.t Y. Let x be any point in E; we have to show that x is an interior point of E in the metric space $(Y, d_{|Y \times Y})$. Since $x \in E$, we know $x \in V$. Since V is open in X, we know that there is a radius $r > 0$ such that $B_{(X,d)}(x, r)$ is contained in V. Strictly speaking, r depends on x, and so we could write r_x instead of r, but for this argument we will only use a single choice of x and so we will not bother to subscript r here. Since $E = V \cap Y$, this means that $B_{(X,d)}(x, r) \cap Y$ is contained in E. But $B_{(X,d)}(x, r) \cap Y$ is exactly the same as $B_{(Y,d_{|Y\times Y})}(x, r)$, and so $B_{(Y,d_{|Y\times Y})}(x, r)$ is contained in E. Thus x is an interior point of E in the metric space $(Y, d_{|Y \times Y})$, as desired. $\square$

§ 1.4 Cauchy sequences and complete metric spaces

Definition 1.8 (Subsequences). Suppose that $(x^{(n)})_{n=m}^{\infty}$ is a sequence of points in a metric space (X, d). Suppose that $n_1, n_2, n_3, \dots$ is an increasing sequence of integers which are at least as large as m, thus

$$m \leq n_1 < n_2 < n_3 < \cdots .$$

Then we call the sequence $(x^{(n_j)})_{j=1}^{\infty}$ a subsequence of the original sequence $(x^{(n)})_{n=m}^{\infty}$.

Lemma 1.2. Let $(x^{(n)})_{n=m}^{\infty}$ be a sequence in (X, d) which converges to some limit x_0. Then every subsequence $(x^{(n_j)})_{j=1}^{\infty}$ of that sequence also converges to x_0.

Definition 1.9 (Limit points). Suppose that $(x^{(n)})_{n=m}^{\infty}$ is a sequence of points in a metric space (X, d), and let $L \in X$. We say that L is a limit point of $(x^{(n)})_{n=m}^{\infty}$ iff for every $N \geq m$ and $\varepsilon > 0$ there exists an $n \geq N$ such that $d(x^{(n)}, L) \leq \varepsilon$.

Definition 1.10 (Cauchy sequences). Let $(x^{(n)})_{n=m}^{\infty}$ be a sequence of points in a metric space (X, d). We say that this sequence is a Cauchy sequence iff for every $\varepsilon > 0$, there exists an $N \geq m$ such that $d(x^{(j)}, x^{(k)}) < \varepsilon$ for all $j, k \geq N$.

Lemma 1.3 (Convergent sequences are Cauchy sequences). Let $(x^{(n)})_{n=m}^{\infty}$ be a sequence in (X, d) which converges to some limit x_0. Then $(x^{(n)})_{n=m}^{\infty}$ is also a Cauchy sequence.

Example. Consider the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, \dots$$

in the metric space $(\mathbb{Q}, d)$ (the rationals $\mathbb{Q}$ with the usual metric $d(x, y) := |x - y|$). While this sequence is convergent in $\mathbb{R}$ (it converges to π), it does not converge in $\mathbb{Q}$ (since $\pi \notin \mathbb{Q}$, and a sequence cannot converge to two different limits).

In certain metric spaces, Cauchy sequences do not necessarily converge. However, if even a part of a Cauchy sequence converges, then the entire Cauchy sequence must converge (to the same limit).

Lemma 1.4. Let $(x^{(n)})_{n=m}^{\infty}$ be a Cauchy sequence in (X, d). Suppose that there is some subsequence $(x^{(n_j)})_{j=1}^{\infty}$ of this sequence which converges to a limit x_0 in X. Then the original sequence $(x^{(n)})_{n=m}^{\infty}$ also converges to x_0.

Definition 1.11 (Complete metric spaces). A metric space (X, d) is said to be complete iff every Cauchy sequence in (X, d) is infact convergent in (X, d).

Proposition 1.6.

1. Let (X, d) be a metric space, and let $(Y, d_{|Y \times Y})$ be a subspace of (X, d). If $(Y, d_{|Y \times Y})$ is complete, then Y must be closed in X.
2. Conversely, suppose that (X, d) is a complete metric space, and Y is a closed subset of X. Then the subspace $(Y, d_{|Y \times Y})$ is also complete.

§ 1.5 Compact metric spaces

Definition 1.12 (Compactness)**.** A metric space (X, d) is said to be compact iff every sequence in (X, d) has at least one convergent subsequence. A subset Y of a metric space X is said to be compact if the subspace $(Y, d_{|Y \times Y})$ is compact.

Definition 1.13 (Bounded sets)**.** Let (X, d) be a metric space, and let Y be a subset of X. We say that Y is bounded iff there exists a ball $B(x, r)$ in X which contains Y.

Proposition 1.7. Let (X, d) be a compact metric space. Then (X, d) is both complete and bounded.

Corollary 1.2 (Compact sets are closed and bounded)**.** Let (X, d) be a metric space, and let Y be a compact subset of X. Then Y is closed and bounded.

Theorem 1.1 (Heine-Borel theorem)**.** Let $(\mathbb{R}^n, d)$ be a Euclidean space with either the Euclidean metric, the taxicab metric, or the sup norm metric. Let E be a subset of $\mathbb{R}^n$. Then E is compact if and only if it is closed and bounded.

The Heine-Borel theorem is not true for more general metrics. For instance, the integers $\mathbb{Z}$ with the discrete metric is closed (indeed, it is complete) and bounded, but not compact, since the sequence $1, 2, 3, 4, \ldots$ is in $\mathbb{Z}$ but has no convergent subsequence.

Theorem 1.2. Let (X, d) be a metric space, and let Y be a compact subset of X. Let $(V_\alpha)_{\alpha \in I}$ be a collection of open sets in X, and suppose that

$$Y \subseteq \bigcup_{\alpha \in I} V_\alpha.$$

Then there exists a finite subset F of I such that

$$Y \subseteq \bigcup_{\alpha \in F} V_\alpha.$$

Proof. We assume for the sake of contradiction that there does not exist any finite subset F of A for which $Y \subset \bigcup_{\alpha \in F} V_\alpha$.

Let y be any element of Y. Then y must lie in at least one of the sets V_α. Since each V_α is open, there must therefore be an $r > 0$ such that $B_{(X,d)}(y, r) \subseteq V_\alpha$. Now let $r(y)$ denote the quantity

$$r(y) := \sup\{r \in (0, \infty) \mid B_{(X,d)}(y, r) \subseteq V_\alpha \text{ for some } \alpha \in A\}.$$

By the above discussion, we know that $r(y) > 0$ for all $y \in Y$. Now, let r_0 denote the quantity

$$r_0 := \inf\{r(y) \mid y \in Y\}.$$

Since $r(y) > 0$ for all $y \in Y$, we have $r_0 \geq 0$. There are two cases: $r_0 = 0$ and $r_0 > 0$.

- **Case 1: $r_0 = 0$.** Then for every integer $n \geq 1$, there is at least one point y in Y such that $r(y) < 1/n$. We thus choose, for each $n \geq 1$, a point $y^{(n)}$ in Y such that $r(y^{(n)}) < 1/n$ (we can do this because of the axiom of choice). In particular we have $\lim_{n\to\infty} r(y^{(n)}) = 0$, by the squeeze test. The sequence $(y^{(n)})_{n=1}^{\infty}$ is a sequence in Y; since Y is compact, we can thus find a subsequence $(y^{(n_j)})_{j=1}^{\infty}$ which converges to a point $y_0 \in Y$.

  As before, we know that there exists some $\alpha \in A$ such that $y_0 \in V_\alpha$, and hence (since V_α is open) there exists some $\varepsilon > 0$ such that $B(y_0, \varepsilon) \subseteq V_\alpha$. Since $y^{(n_j)}$ converges to 0, there must exist an $N \geq 1$ such that $y^{(n_j)} \in B(y_0, \varepsilon/2)$ for all $n \geq N$. In particular, by the triangle inequality we have $B(y^{(n_j)}, \varepsilon/2) \subseteq B(y_0, \varepsilon)$, and thus $B(y^{(n_j)}, \varepsilon/2) \subseteq V_\alpha$. By definition of $r(y^{(n_j)})$, this implies that $r(y^{(n_j)}) \geq \varepsilon/2$ for all $n \geq N$. But this contradicts the fact that $\lim_{n\to\infty} r(y^{(n)}) = 0$.

- **Case 2: $r_0 > 0$.** In this case we now have $r(y) > r_0/2$ for all $y \in Y$. This implies that for every $y \in Y$ there exists an $\alpha \in A$ such that $B(y, r_0/2) \in V_\alpha$.

  We now construct a sequence $y^{(1)}, y^{(2)}, \ldots$ by the following recursive procedure. We let $y^{(1)}$ be any point in Y. The ball $B(y^{(1)}, r_0/2)$ is contained in one of the V_α and thus cannot cover all of Y, since then we would then obtain a finite cover, a contradiction. Thus there exists a point $y^{(2)}$ which does not lie in $B(y^{(1)}, r_0/2)$, so in particular $d(y^{(2)}, y^{(1)}) \geq r_0/2$. Choose such a point $y^{(2)}$. The set $B(y^{(1)}, r_0/2) \cup B(y^{(2)}, r_0/2)$ cannot cover all of Y, since we would then obtain two sets V_{α_1} and V_{α_2} which covered Y, a contradiction again. So we can choose a point $y^{(3)}$ which does not lie in $B(y^{(1)}, r_0/2) \cup B(y^{(2)}, r_0/2)$, so in particular $d(y^{(3)}, y^{(1)}) \geq r_0/2$ and $d(y^{(3)}, y^{(2)}) \geq r_0/2$. Continuing in this fashion we obtain a sequence $(y^{(n)})_{n=1}^{\infty}$ in Y with the property that $d(y^{(k)}, y^{(j)}) \geq r_0/2$ for all $k > j$. In particular the sequence $(y^{(n)})_{n=1}^{\infty}$ is not a Cauchy sequence, and in fact no subsequence of $(y^{(n)})_{n=1}^{\infty}$ can be a Cauchy sequence either. But this contradicts the assumption that Y is compact.

$\square$

2
Continuous functions on metric spaces

§ 2.1 Continuous functions

Definition 2.1 (Continuous functions). Let (X, d_X) be a metric space, and let (Y, d_Y) be another metric space, and let $f : X \to Y$ be a function. If $x_0 \in X$, we say that f is continuous at x_0 iff for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $d_Y(f(x), f(x_0)) < \varepsilon$ whenever $d_X(x, x_0) < \delta$. We say that f is continuous iff it is continuous at every point $x \in X$.

Remark. Continuous functions are also sometimes called continuous maps. Mathematically, there is no distinction between the two terminologies.

Remark. If $f : X \to Y$ is continuous, and K is any subset of X, then the restriction $f|_K : K \to Y$ of f to K is also continuous.

Theorem 2.1 (Continuity preserves convergence). Suppose that (X, d_X) and (Y, d_Y) are metric spaces. Let $f : X \to Y$ be a function, and let $x_0 \in X$ be a point in X. Then the following three statements are logically equivalent:

1. f is continuous at x_0.
2. Whenever $(x^{(n)})_{n=m}^{\infty}$ is a sequence in X which converges to x_0 w.r.t the metric d_X, the sequence $(f(x^{(n)}))_{n=1}^{\infty}$ converges to $f(x_0)$ w.r.t the metric d_Y.
3. For every open set $V \subset Y$ that contains $f(x_0)$, there exists an open set $U \subset X$ containing x_0 such that $f(U) \subseteq V$.

Theorem 2.2. Let (X, d_X) be a metric space, and let (Y, d_Y) be another metric space. Let $f : X \to Y$ be a function. Then the following four statements are equivalent:

1. f is continuous.
2. Whenever $(x^{(n)})_{n=1}^{\infty}$ is a sequence in X which converges to some point $x_0 \in X$ w.r.t the metric d_X, the sequence $(f(x^{(n)}))_{n=1}^{\infty}$ converges to $f(x_0)$ w.r.t the metric d_Y.
3. Whenever V is an open set in Y, the set $f^{-1}(V) := \{x \in X \mid f(x) \in V\}$ is an open set in X.
4. Whenever F is a closed set in Y, the set $f^{-1}(F) := \{x \in X \mid f(x) \in F\}$ is a closed set in X.

Corollary 2.1 (Continuity preserved by composition). Let $(X, d_X), (Y, d_Y)$, and (Z, d_Z) be metric spaces.

1. If $f : X \to Y$ is continuous at a point $x_0 \in X$, and $g : Y \to Z$ is continuous at $f(x_0)$, then the composition $g \circ f : X \to Z$, defined by $g \circ f(x) := g(f(x))$, is continuous at x_0.

2. If $f : X \to Y$ is continuous, and $g : Y \to Z$ is continuous, then $g \circ f : X \to Z$ is also continuous.

Example. If $f : X \to \mathbb{R}$ is a continuous function, then the function $f^2 : X \to \mathbb{R}$ defined by $f^2(x) := f(x)^2$ is automatically continuous also. This is because we have $f^2 = g \circ f$, were $g : \mathbb{R} \to \mathbb{R}$ is the squaring function $g(x) := x^2$, and g is a continuous function.

§ 2.2 Continuity and product spaces

Given two functions $f : X \to Y$ and $G : X \to Z$, one can define their direct sum $f \oplus g : X \to Y \times Z$ defined by $f \oplus g(x) := (f(x), g(x))$, i.e., this is the function taking values in the Cartesian product $Y \times Z$ whose first co-ordinate is $f(x)$ and whose second co-ordinate is $g(x)$. For instance, if $f : \mathbb{R} \to \mathbb{R}$ is the function $f(x) := x^2 + 3$, and $g : \mathbb{R} \to \mathbb{R}$ is the function $g(x) = 4x$, then $f \oplus g : \mathbb{R} \to \mathbb{R}^2$ is the function $f \oplus g(x) := (x^2 + 3, 4x)$. The direct sum operation preserves continuity:

Lemma 2.1. Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions, and let $f \oplus g : X \to \mathbb{R}^2$ be their direct sum. We give $\mathbb{R}^2$ the Euclidean metric.

1. If $x_0 \in X$, then f and g are both continuous functions at x_0 iff $f \oplus g$ is continuous at x_0.

2. f and g are both continuous iff $f \oplus g$ is continuous.

Lemma 2.2. The addition function $(x, y) \mapsto x + y$, the subtraction function $(x, y) \mapsto x - y$, the multiplication function $(x, y) \mapsto xy$, the maximum function $(x, y) \mapsto \max(x, y)$, and the minimum function $(x, y) \mapsto \min(x, y)$, are all continuous functions from $\mathbb{R}^2 \to \mathbb{R}$. The division function $(x, y) \mapsto x/y$ is a continuous function from $\mathbb{R} \times \{\mathbb{R} \backslash \{0\}\} = \{(x, y) \in \mathbb{R}^2 \mid y \neq 0\} \to \mathbb{R}$. For any real number c, the function $x \mapsto cx$ is a continuous function from $\mathbb{R} \to \mathbb{R}$.

Combining these lemmas we obtain

Corollary 2.2. Let (X, d) be a metric space, let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be functions. Let c be a real number.

1. If $x_0 \in X$ and f and g are continuous at x_0, then the functions $f + g : X \to \mathbb{R}$, $f - g : X \to \mathbb{R}$, $fg : X \to \mathbb{R}$, $\max(f, g) : X \to \mathbb{R}$, $\min(f, g) : X \to \mathbb{R}$, and $cf : X \to \mathbb{R}$ are also continuous at x_0. If $g(x) \neq 0$ for all $x \in X$, then $f/g : X \to \mathbb{R}$ is also continuous at x_0.
2. If f and g are continuous, then the functions $f + g : X \to \mathbb{R}$, $f - g : X \to \mathbb{R}$, $fg : X \to \mathbb{R}$, $\max(f, g) : X \to \mathbb{R}$, $\min(f, g) : X \to \mathbb{R}$, and $cf : X \to \mathbb{R}$ are also continuous at x_0. If $g(x) \neq 0$ for all $x \in X$, then $f/g : X \to \mathbb{R}$ is also continuous at x_0.

Proof. We first prove (1). Since f and g are continuous at x_0, then by Lemma 2.1 $f \oplus g : X \to \mathbb{R}^2$ is also continuous at x_0. On the other hand, from Lemma 2.2 the function $(x, y) \mapsto x + y$ is continuous at every point in $\mathbb{R}^2$, and in particular is continuous at $f \oplus g(x_0)$. If we then compose these functions using Corollary 2.1 we conclude that $f + g : X \to \mathbb{R}$ is continuous. A similar argument gives the continuity of $f - g, fg, \max(f, g), \min(f, g)$ and cf. To prove the claim for f/g, we restrict the range of g from $\mathbb{R} \to \mathbb{R} \backslash \{0\}$, and then one can argue as before. The claim (2) follows immediately from (1). □

§ 2.3 Continuity and compactness

Theorem 2.3 (Continuous maps preserve compactness)**.** Let $f : X \to Y$ be a continuous map from one metric space (X, d_X) to another (Y, d_Y). Let $K \subseteq X$ be any compact subset of X. Then the image $f(K) := \{f(x) \mid x \in K\}$ of K is also compact.

Proposition 2.1 (Maximum principle)**.** Let (X, d) be a compact metric space, and let $f : X \to \mathbb{R}$ be a continuous function. Then f is bounded. Furthermore, f attains its maximum at some point $x_{\max} \in X$, and also attains its minimum at some point $x_{\min} \in X$.

Remark. This principle can fail if X is not compact.

Another advantage of continuous functions on compact sets is that they are uniformly continuous.

Definition 2.2 (Uniform continuity)**.** Let $f : X \to Y$ be a map from one metric space (X, d_X) to another (Y, d_Y). We say that f is uniformly continuous if, for every $\varepsilon > 0$, there exists a $\delta > 0$ such that $d_Y(f(x), f(x')) < \varepsilon$ whenever $x, x' \in X$ are such that $d_X(x, x') < \delta$.

Every uniformly continuous function is continuous, but not conversely.

Theorem 2.4. Let (X, d_X) and (Y, d_Y) be metric spaces, and suppose that (X, d_X) is compact. If $f : X \to Y$ is a function, then f is continuous if and only if it is uniformly continuous.

Proof. If f is uniformly continuous then it is also continuous. Now suppose that f is continuous. Fix $\varepsilon > 0$. For every $x_0 \in X$, the function f is continuous at x_0. Thus there exists a $\delta(x_0) > 0$, depending on x_0, such that $d_Y(f(x), f(x_0)) < \varepsilon/2$ whenever $d_X(x, x_0) < \delta(x_0)$. In particular, by the triangle inequality this implies that $d_Y(f(x), f(x')) < \varepsilon$ whenever $x \in B_{(X,d_X)}(x_0, \delta(x_0)/2)$ and $d_X(x', x) < \delta(x_0)/2$.

Now consider the (possibly infinite) collection of balls

$$\{B_{(X,d_X)}(x_0, \delta(x_0)/2) \mid x_0 \in X\}.$$

Each ball in this collection is of course open, and the union of all these balls covers X, since each point $x_0 \in X$ is contained in its own ball $B_{(X,d_X)}(x_0, \delta(x_0)/2)$. Hence, there exist a finite number of points $x_1, \ldots, x_n$ such that the balls $B_{(X,d_X)}(x_j, \delta(x_j)/2)$ for $j = 1, \ldots, n$ cover X:

$$X \subseteq \bigcup_{j=1}^{n} B_{(X,d_X)}(x_j, \delta(x_j)/2).$$

Now let $\delta := \min_{j=1}^{n} \delta(x_j)/2$. Since each of the $\delta(x_j)$ are positive, and there are only a finite number of j, we see that $\delta > 0$. Now let x, x' be any two points in X such that $d_X(x, x') < \delta$. Since the balls $B_{(X,d_X)}(x_j, \delta(x_j)/2)$ cover X, we see that there must exist $1 \leq j \leq n$ such that $x \in B_{(X,d_X)}(x_j, \delta(x_j)/2)$. Since $d_X(x, x') < \delta$, we have $d_X(x, x') < \delta(x_j)/2$, and so by the previous discussion we have $d_Y(f(x), f(x')) < \varepsilon$. We have thus found a δ such that $d_Y(f(x), f(x')) < \varepsilon$ whenever $d(x, x') < \delta$, and this proves uniform continuity as desired. $\square$